

Mathematical Formulae Used

Measure	Formula	Interpretation in CCD Sales Context
Mean ($\bar{x}$)	$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$	Average sales per category, helps understand typical revenue.
Median	Middle value of sorted x_i	Shows central tendency, less influenced by outliers.
Standard Deviation (σ)	$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2}$	Measures sales variability across categories.
Coefficient of Variation (CV)	$CV = \frac{\sigma}{\bar{x}} \times 100$	Compares relative variability of sales across product types.
Correlation (r)	$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$	Measures relationship between food and beverage sales.
Mean Absolute Error (MAE)	$MAE = \frac{1}{N} \sum_{i=1}^N y_i - \hat{y}_i $	Average absolute difference between predicted and actual sales.
Mean Absolute Percentage Error (MAPE)	$MAPE = \frac{100}{N} \sum_{i=1}^N \left \frac{y_i - \hat{y}_i}{y_i} \right $	Prediction error expressed as a percentage, easier to interpret.
Coefficient of Determination (R^2)	$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$	Explains how much variance in sales is explained by the model.

Summary: These measures were chosen to provide a complete statistical picture of CCD sales. Central tendency (Mean, Median) highlighted typical revenue, variability measures (Standard Deviation, CV) showed spread, correlation checked relationships, and forecasting metrics (MAE, MAPE, R^2) validated model accuracy. Together, they ensured both reliable analysis and actionable insights.